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Lab 5 Report

1. Converting the raw audio waveform into time-frequency features in a spectrogram helps compress long sequences of samples into a compact, 2D representation of the frequencies present. Spectrograms capture what frequencies are present at which times, which is the fingerprint used to distinguish words from one another, while a time-domain representation of the waveform displays redundant information sensitive to phase shifts.
2. It is important to have a representative dataset when creating an INT8 TensorFlow Lite model because in full integer quantization, weights and activations are converted from 32-bit floats to 8-bit integers. The representative dataset is a small sample of data that is run through the model for the quantizer to observe and record the range of values for each layer's activations during inference which exposes the quantizer to the activation ranges instead of the weight ranges alone.
3. The audio_processor.get_data() loads audio clips from the dataset, preprocesses it by partitioning data into training, testing, and validation, and then computing the spectrogram of MFCC features from the raw waveform. Importantly, this function returns a structure containing the file paths and labels for each of these partitions (train, test, validate) for model training or evaluation.
4. G_model is an array of bytes that stores the model weights and metadata that sits in the microcontroller's flash memory. G_model_len is the total number of bytes in that array (the model's size) which TensorFlow Lite's needs to know before loading it. These must be correctly copied to the Arduino source file because the microcontroller runs on the program's exact binary instead of running a file at runtime. Any corruption or truncation of the byte array can cause faulty model behavior.
5.

Message (Enter to send message to 'Arduino Nano 33 BLE' on 'C

```
15:41:27.869 -> Heard unknown (208) @4576ms
15:41:33.011 -> Heard yes (210) @9696ms
15:41:34.694 -> Heard unknown (206) @11408ms
15:41:36.888 -> Heard unknown (201) @13600ms
15:41:43.327 -> Heard unknown (201) @19984ms
15:41:46.145 -> Heard unknown (209) @22848ms
15:41:48.380 -> Heard yes (201) @25040ms
15:41:51.941 -> Heard unknown (201) @28640ms
15:41:54.312 -> Heard unknown (204) @31024ms
15:41:58.916 -> Heard unknown (205) @35584ms
15:42:02.644 -> Heard unknown (204) @39312ms
15:42:06.301 -> Heard no (207) @42960ms
15:42:12.637 -> Heard unknown (202) @49296ms
15:42:15.728 -> Heard unknown (201) @52416ms
```